WEEK-2

Independence of Random Variables

For random var. x and y are independent if

$$P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$$

for continuous var. $\rightarrow$ $f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$

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i.i.d (Independent Indentically distributed)

$$P(X_1 > j, X_2 > j, \ldots, X_n > j) = [P(X > j)]^n$$

- Independent
- Same distribution

if $X \sim \text{Geometric}(p)$

Q.1 Let $X_1, X_2, X_3, \ldots, X_{10} \sim$ i.i.d. Geometric $\left(\frac{1}{5}\right)$. Find the Probability that $(X_1 \geq 10, X_2 > 10, \ldots X_{10} > 10)$.

Sol. • Geometric Distr: $= P(X > k) = (1-p)^k$

$= (0.8)^{10}$

• Use Independence: $= P(X_1 > 10, \ldots, X_{10} > 10)$

$$\Rightarrow \prod_{i=1}^{10} P(X_i > 10) = (0.8^{10})^{10} = 0.8^{100}$$

Q.2 Let $X \sim$ Geometric (p) and

$$f(x) = \begin{cases} x/2 & \rightarrow \text{if } x \text{ is an even no.} \\ \frac{x+1}{2} & \rightarrow \text{if } x \text{ is an odd no.} \end{cases}$$

Find the probability that $f(X) = k$ where k is in range of $f(x)$.

Sol.

$$f(x) = \frac{x}{2} \Big] \rightarrow \text{even}$$

$$f(x) = \frac{x+1}{2} \Big] \rightarrow \text{odd}$$

- $k = \frac{x}{2}$ $\boxed{x = 2k}$
- $k = \frac{x+1}{2}$ $\boxed{x = 2k-1}$

$$P(f(x) = x) = p \cdot (1-p)^{x-1}$$

$$P_{net} = P_{even} + P_{odd}$$

$$= p \cdot (1-p)^{2k-1} + p \cdot (1-p)^{(2k-1)-1}$$

$$= p\,(1-p)^{2k-2}\left[(1-p) + 1\right]$$

Ans. $\Rightarrow$ $p \cdot (1-p)^{2(k-1)} \cdot [2-p]$

Functions of Multiple Random Variables

Q.1. A random experiment consists of rolling an unbiased die 2 times. Let X denote the no. obtained on the first die and Y denote the number obtained on the second die. The joint PMF of X and Y is denoted by $f_{XY}(t_1, t_2)$. Let $Z = X + Y$. Choose the correct statements from the following.

Sol.

- $P(Z=3) = \sum_{x=1}^{2} f_{XY}(x, 3-x)$ ✓

  $P(Z=3) = f_{XY}(1,2) + f_{XY}(2,1)$

- # $\boxed{\text{Convulation formula for sum of independent variables}}$

  $P(Z=z) = \sum P(X=x) \cdot P(Y = z-x)$ → works because X and Y are independent

  $$\boxed{f_Z(z) = \sum_{x=1}^{6} f_X(x) \cdot f_Y(z-x)}$$ ✓

Q.2 # Poisson + Conditional

Let the random variables X and Y, which represent the no. of people visiting shopping malls in C-1 and C-2 in 1hr interval, respectively, follow the Poisson distribution. The avg. no. of people visiting the shopping malls in C-1 and C-2 is 10/hr and 20/hr, respectively. Assume X and Y are independent. Let Z denote the total no. of people visiting shopping malls in C-1 and C-2. Find the pmf of Z, $f_Z(z)$.

Sol.

$\lambda_Z = \lambda_X + \lambda_Y$
$= 10 + 20$
$\Rightarrow 30$

$f_Z(z) = \frac{\lambda_z^z \cdot e^{-\lambda_z}}{z!}$ $\quad f_X(x) = \frac{\lambda_x^x \cdot e^{-\lambda_x}}{x!}$

$\Rightarrow \frac{(30)^z \cdot e^{-30}}{z!}$

$\boxed{\lambda_X = 10, \ \lambda_Y = 20}$

Q. Find the conditional distribution of Y given that the total no. of people visiting shopping malls in C-1 and C-2 is 30.

Sol.

$X \sim \text{Poisson}(10)$
$Y \sim \text{Poisson}(20)$
independent: $Z = X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$

$Y \mid X + Y = n) \sim \text{Binomial}\left(n, \frac{\lambda_1}{\lambda_1 + \lambda_2}\right)$
$= \text{Binomial}\left(30, \frac{2}{3}\right)$

Min-Max Random Variables

- $X_1, X_2, \ldots, X_{10} \sim \text{Binomial}\left(6, \frac{1}{2}\right)$ i.i.d.
- $Z_1 = \max(X_1, \ldots, X_{10})$
- $Z_2 = \min(X_1, \ldots, X_{10})$

Q. Calculate $P(Z_1 \le 2)$

Key Formulas:

$P(\min \ge z) = [P(X \ge z)]^n$
$P(\max \le z) = [P(X \le z)]^n$

Sol Since i.i.d,

$P(Z_i \le 2) = P(X_1 \le 2)^{10}$

Step:1 Compute $P(X \le 2)$

$X \sim \text{Binomial}\left(6, \frac{1}{2}\right)$

$P(X \le 2) = \sum_{k=0}^{6} \binom{6}{k} \left(\frac{1}{2}\right)^6 = \frac{1}{2^6}\left[\binom{6}{0} + \binom{6}{1} + \binom{6}{2}\right]$

$= \frac{22}{64} = \frac{22}{2^6}$

Final ans $\Rightarrow \left(\frac{22}{2^6}\right)^{10}$

Q. Calculate $F_{Z_2}(2)$. $\longrightarrow = [P(Z_2 > 2)]^{10}$

Sol.

$= 1 - P(Z_2 \le 2)$

$Z_2 = \min(X_i)$

$P(Z_2 > 2) = 1 - P(\text{all } X_i \le 2)$

$\Rightarrow \boxed{1 - \left(\frac{21}{25}\right)^{10}}$

Geometric + Min

$\min(X, Y) \sim Geo(P_1 + P_2 - P_1P_2)$

Q. Let $X \sim$ Geometric $\left(\frac{1}{8}\right)$ and $Y \sim$ Geometric $\left(\frac{5}{8}\right)$ be independent. Define $Z = \min(X, Y)$. Find the distribution of Z.

Sol. $P_Z = 1 - (1 - P_X)(1 - P_Y) \longrightarrow$ ☆ (Shorcut Formula if)

$= 1 - \left(1 - \frac{1}{8}\right)\left(1 - \frac{5}{8}\right) = 1 - \left(\frac{5}{8}\right)\left(\frac{1}{8}\right) \Rightarrow 1 - \frac{5}{8} = \boxed{\frac{31}{36}}$

Function of 2 Random Variable

CDF (Cumulative Distributive Function)

$\hookrightarrow F_X(x) = P(X \le x)$

Q. Let X and Y be 2 independent random variables such that $X \sim$ Bernoulli(0.2) and $Y \sim$ Bernouli(0.4). Let another random var. Z be defined as $Z = X + Y$. Find the value of $F_{X|Z=1}(1)$.

Sol. $P(X = 1 \mid Z = 1) = \dfrac{P(X=1, Z=1)}{P(Z=1)} \Rightarrow \dfrac{P(X=1, Y=0)}{P(X+Y=1)}$

$X + Y = 1$: $1 + 0$, $0 + 1$

$(0.2)(0.6)$

$P(1,0) + P(0,1)$

$= (0.2)(0.6) + (0.8)(0.4) = 0.44$

$P(\text{ans}) \Rightarrow \dfrac{0.12}{0.44} = \dfrac{3}{11} \approx 0.27$

Q. Let $X \sim$ Geometric(P) and $Y \sim$ Geometric(P) be independent and let $Z = X + Y$. Determine the values of P for which $P(Z = 14) < P(Z = 15)$.

Sol. $X + Y = K$

no. of ways $= K - 1$

$P \cdot (1-P)^{K-1}$

$14 \rightarrow 13$ ways

$15 \rightarrow 14$ ways

$P \cdot (1-P)^{13} < 14 \cdot (1-P)^{13}$

$P < 0.071$ $\quad 14P < 1$ $\quad P < \frac{1}{14} \rightarrow 0.071$

WEEK-3

Q. ① Let X_1, X_2 and X_3 be 3 independent and identically distributed Poisson random variables with $\lambda_i = 2$ for all i. Find the prob. that exactly one of the X_i equals 0.

~~Enter the an~~

Sol. ① $P(X=0) = e^{-2}$

$P(X=2) = \frac{2^2 \cdot e^{-2}}{2!} = 2e^{-2}$

$P(X \neq 0, 2) = 1 - P(0) - P(2) = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2}$

② no. of ways $= 3! =$ 6 ways

③ in one arrangement $= (2e^{-2}) \cdot (e^{-2}) \cdot (1-3e^{-2})$

$= \boxed{2e^{-4}(1-3e^{-2}) \times 6}$

all possible ways

Q ② Let the rand. v.r. X and Y, in a 1-hr interval follow poisson distrib.

X → no. of calls (A), Y → no. of calls (B)

The avg. no. of call $A = 2/hr$, $B = 1/hr$

~~Assu~~. Assume X and Y are independent. If Z denotes the total no. of calls in A and B, find $f_{Y|Z=4}(3)$.

Sol. $X \sim \text{Poisson}(\lambda_1)$

$Y \sim \text{Poisson}(\lambda_2)$

$Z = X+Y \sim \text{Poisson}(\lambda_1+\lambda_2)$

$Y \mid (X+Y=n) \sim \text{Binomial}\left(n, \frac{\lambda_2}{\lambda_1+\lambda_2}\right)$

$\sim \text{Bino}\left(4, \frac{1}{3}\right)$

$P(Y=3 \mid Z=4) = {}^4C_3 \left(\frac{1}{3}\right)^3 \cdot \left(\frac{2}{3}\right)^1$

$= \frac{8}{81}$